Exercise Sheet 2

1. Show that a linear filter (a_j) passes an arbitrary polynomial of degree k without distortion, i.e.
$$m_t = \sum_j a_j m_{t-j}$$
for all k-th degree polynomials $m_t = c_0 + c_1 t + \cdots + c_k t^k$, if and only if (a) $\sum_j a_j = 1$ and (b) $\sum_j j^r a_j = 0$, for $r = 1, \ldots, k$.

2. Assume that $\{Y_t | t \in \mathbb{Z}\}$ is weakly stationary. Define
$$Q_k(B) = c_0 + c_1 B + \cdots + c_k B^k$$
as "backshift polynomial". Show that
$$X_t = Q_k(B) Y_t = c_0 Y_t + c_1 Y_{t-1} + \cdots + c_k Y_{t-k}$$
is again weakly stationary.

3. Show that if $P_n(B)$ is another backshift polynomial of degree n, then
$$Q_k(B)\Big[P_n(B) Y_t\Big] = R_{n+k}(B) Y_t$$
with $R_{n+k}(z) = Q_k(t) P_n(z)$.

4. In this exercise we derive the mean and the ACF of an AR(1) process
$$X_t = \alpha X_{t-1} + Z_t, \tag{1}$$
where $\{Z_t | \, t \in \mathbb{Z}\}$ is an i.i.d. sequence with $EZ_t = 0$ and $\mathrm{Var}(Z_t) = \sigma_Z^2 < \infty$. The target is **to avoid the expansion** of X_t into an MA(∞) process. Instead, we only use the following assumptions: (A) There is a stationary solution to equation (1). (B) It holds that $\mathrm{Cov}(Z_t, X_s) = 0$ for all $s < t$. Show that such a solution cannot exist if $|\alpha| \geq 1$.

5. Assume that $\kappa(\cdot)$ is non-negative definite and even. Show (directly, i.e. without using that it is an ACF) that $\kappa(0) \geq 0$ and $\kappa(0) \geq |\kappa(h)|$.

6. Compute the longrun variance of the MA(1) process $X_t = Z_t + \theta Z_{t-1}$ where $\{Z_t\}$ is i.i.d. with $EZ_t = 0$ and $EZ_t^2 = 1$. What happens if $\theta = -1$?

7. Compute the longrun variance of the AR(1) process defined in (1). Discuss the cases $\alpha \to 1$, $\alpha \to -1$ and $\alpha = 0$.

8. If $\{X_t\}$ is a stationary time series with mean μ then the usual estimator for μ is the sample mean $\hat{\mu} = \frac{1}{n}(X_1 + \cdots + X_n)$. We have shown for this estimator that under mild regularity assumptions the law of large numbers will hold, i.e. $\hat{\mu} \to \mu$ in probability. Also, we know that this estimator is unbiased, i.e. $E\hat{\mu} = \mu$. Assume we have $X_t = \mu + Z_t + \frac{1}{2} Z_{t-1}$, where $\{Z_t\} \sim \mathrm{WN}(0, 1)$ and consider a sample X_1, X_2, X_3. Can you find an unbiased estimator for μ of the form $\tilde{\mu} = a_1 X_1 + a_2 X_2 + a_3 X_3$ such that $\mathrm{Var}(\tilde{\mu}) < \mathrm{Var}(\hat{\mu})$?